MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)
REGULAR EXAMINATION October 2026
Fifth SEMESTER
Skill Enhancement Course (SEC) - MG5SECFSC300 - RESEARCH METHODOLOGY AND STATISTICS
(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION / EVALUATION

Duration: 1 hour 30 minutes
Maximum Marks: 50

General Instructions for Examiners:

1. Award marks for equivalent scientifically and methodologically correct answers and statistical definitions.
2. The value points below are indicative, not exhaustive, structured strictly in line with the syllabus and Bloom's Taxonomy levels indicated in the question paper.
3. For questions with multiple parts or sub-points, award marks strictly in accordance with the allocated value points.
4. Under no circumstances should marks awarded exceed the maximum mark allotted for each question or section.
5. Students are instructed to attempt at least one question from each Course Outcome (CO). Apply university examination regulations regarding choice patterns and course outcome attainability.

Part A – Short Answer Type Questions

Answer any 10 questions. Each question carries 2 marks. Maximum: 20 Marks.

Question 1. What is the primary purpose of applied research? (2 Marks) [CO1, U]

- Definition of applied research: Finding practical solutions to immediate, specific, and real-world problems facing an individual, organization, or society (as opposed to pure/fundamental research aiming at theoretical expansion) **— 1 mark**
- Application/Example: Applying scientific principles to develop practical tools, policy solutions, forensic protocols, or decision-making frameworks **— 1 mark**

Question 2. Analyse the essential characteristics of a good sample with reference to representativeness, adequacy, and freedom from bias. (2 Marks) [CO3, An]

- Representativeness and Adequacy: The sample must accurately reflect the proportional characteristics, stratification, and diversity of the target population; the sample size (n) must be sufficiently large to provide statistical power and minimize sampling error **— 1 mark**
- Freedom from Bias: Every member of the population should have a known or uncontrolled chance of inclusion, avoiding systematic distortion, self-selection bias, or investigator prejudice **— 1 mark**

Question 3. How is class width determined? (2 Marks) [CO4, U]

- Formula / Definition: Class width (interval size, h or c) is determined by dividing the total range of the dataset by the desired number of class intervals: Class Width = (Upper Class Limit - Lower Class Limit) or Range / Number of Classes (k) **— 1 mark**
- Empirical guideline: Calculated using Sturges' Rule for number of classes ($k = 1 + 3.322 \log_{10} N$) or tailored to maintain uniform, non-overlapping, and continuous bins across the data distribution **— 1 mark**

Question 4. What is data collection? (2 Marks) [CO2, K]

- Definition: The systematic and organized process of gathering, measuring, and recording information or observations on relevant variables in an established systematic fashion **— 1 mark**
- Objective/Relevance: To enable a researcher to test empirical hypotheses, answer stated research questions, evaluate outcomes, and draw valid scientific conclusions **— 1 mark**

Question 5. Differentiate between convenience sampling and snowball sampling based on participant recruitment. (2 Marks) [CO3, An]

- Convenience Sampling: Participants are recruited solely based on ease of access, geographic proximity, or ready availability to the researcher (e.g., passersby, available volunteers) **— 1 mark**
- Snowball Sampling: Initial identified participants who meet specific research criteria serve as informants to refer, recruit, or nominate subsequent participants from their network (ideal for hidden, hard-to-reach, or specialized populations) **— 1 mark**

Question 6. Mention two characteristics of conceptual research. (2 Marks) [CO1, U]

- Focus on ideas and theories: Relies on abstract theories, intellectual concepts, and philosophical frameworks rather than conducting empirical experiments or practical field trials **— 1 mark**
- Re-interpretation and synthesis: Involves critical re-examination, re-interpretation, refinement, or synthesis of existing theories and concepts to develop new conceptual paradigms or models **— 1 mark**

Question 7. Examine the major factors that influence the selection of a sampling design. (2 Marks) [CO3, An]

- Nature and structure of the population: Population size, homogeneity vs. heterogeneity, geographic dispersion, and accessibility of a complete sampling frame **— 1 mark**
- Resource constraints and precision: Budget, time limits, available manpower, and the desired level of precision, confidence interval, and acceptable margin of error **— 1 mark**

Question 8. What is grouped data? (2 Marks) [CO4, U]

- Definition: Raw data that has been organized, aggregated, and classified into mutually exclusive and exhaustive categories, frequency bins, or class intervals **— 1 mark**

- Characteristics: Presented as a frequency distribution table to simplify large datasets and reveal underlying patterns, though individual raw data values are condensed **— 1 mark**

Question 9. How is coefficient of variation used to compare variability? (2 Marks) [CO5, U]

- Definition and Formula: Relative measure of dispersion defined as the ratio of standard deviation (σ or s) to the arithmetic mean (μ or $\bar{x}$), expressed as a percentage: CV = (s / $\bar{x}$) * 100 **— 1 mark**
- Application: Used to compare the degree of variation, stability, or consistency between two or more datasets measured in different units or having widely different magnitudes of means (lower CV indicates higher consistency/uniformity) **— 1 mark**

Question 10. What is the importance of objectivity in research? (2 Marks) [CO1, K]

- Elimination of bias: Ensures that observations, data collection, analytical findings, and conclusions are free from personal prejudice, value judgments, investigator expectations, and emotional bias **— 1 mark**
- Scientific validity and reproducibility: Enhances scientific credibility, transparency, universal acceptance, and reproducibility by other independent researchers across the global scientific community **— 1 mark**

Question 11. Differentiate between population and unit of analysis with a suitable example. (2 Marks) [CO3, An]

- Distinction: 'Population' is the entire aggregated pool or universe of cases/elements that possess specific characteristics of interest. 'Unit of analysis' is the specific individual entity, subject, or object being directly analyzed and measured within that study **— 1 mark**
- Illustrative Example: In a study assessing bloodstain pattern analysis in homicides across Kerala: Population = All registered homicide crime scenes in Kerala; Unit of analysis = An individual bloodstain pattern / individual crime scene docket **— 1 mark**

Question 12. Explain a discrete variable with an example. (2 Marks) [CO4, U]

- Definition: A quantitative variable that can only assume distinct, separate, countable values with no intermediate fractional values between adjacent points (takes isolated integer values) **— 1 mark**
- Example: Number of bullet cartridges recovered from a crime scene, number of ridge bifurcations in a fingerprint, number of PCR amplification cycles (accept any valid countable example) **— 1 mark**

Part B – Short Essay Type Questions

Answer any 5 questions. Each question carries 4 marks. Maximum: 20 Marks.

Question 13. Explain the major steps involved in conducting analytical research. (4 Marks) [CO1, K]

- Formulating research problem and hypothesis: Clearly defining the core question, operationalizing key variables, and postulating testable null and alternative hypotheses **— 1 mark**
- Rigorous research design and data acquisition: Developing controlled experimental setups or selecting validated secondary datasets to ensure measurement reliability and validity **— 1 mark**
- Data processing and critical analytical evaluation: Applying statistical tools, comparative analyses, and inferential tests to examine causal relationships, correlations, and variances **— 1 mark**
- Hypothesis testing, synthesis, and reporting: Interpreting findings against existing literature, drawing objective inferences, acknowledging limitations, and drafting structured research reports **— 1 mark**

Question 14. Analyse the limitations of secondary data and explain the factors that should be examined before using it in research. (4 Marks) [CO3, An]

- Inherent limitations of secondary data: Risk of obsolescence/outdated info, lack of fit with specific research objectives, inaccurate or inconsistent operational definitions, and inability to verify original data collection rigor **— 1 mark**
- Suitability and Relevance factor: Evaluating whether the secondary data's scope, geographic coverage, time horizon, and target population align with the current study's research questions **— 1 mark**
- Reliability and Authenticity of source: Assessing the credibility, institutional reputation, qualifications of the original investigators, and funding agencies of the source dataset **— 1 mark**
- Adequacy and Methodological integrity: Verifying sample size adequacy, sampling techniques utilized, data completeness (missing values), measurement errors, and freedom from commercial/political bias **— 1 mark**

Question 15. Explain interval and ratio scales and distinguish between them. (4 Marks) [CO4, An]

- Interval Scale explanation: A quantitative measurement scale where order and exact differences (intervals) between values are equal and meaningful, but lacks a true/absolute zero point (zero is arbitrary) **— 1 mark**
- Interval examples and operations: Celsius/Fahrenheit temperature scales, calendar dates, IQ scores; allows addition and subtraction, but ratios are meaningless (e.g., 20°C is not twice as hot as 10°C) **— 1 mark**
- Ratio Scale explanation: A quantitative measurement scale possessing all properties of interval scales plus a true, non-arbitrary, absolute zero point denoting total absence of the measured attribute **— 1 mark**
- Ratio examples and comparative distinction: Weight, height, age, blood alcohol concentration (BAC), distance; permits all mathematical and arithmetic operations including multiplication, division, and meaningful ratios (e.g., 80 kg is exactly twice as heavy as 40 kg) **— 1 mark**

Question 16. Explain the steps involved in formulating a research problem. (4 Marks) [CO2, K]

- Statement of broad subject area and domain selection: Identifying a broad topic of interest based on intellectual curiosity, practical forensic needs, societal challenges, or technical gaps **— 1 mark**
- Extensive literature review and gap identification: Comprehensively reviewing academic journals, standard reference texts, and previous findings to identify theoretical controversies, unanswered questions, and methodological blind spots **— 1 mark**
- Dissecting and narrowing down into sub-problems: Deconstructing the broad area into focused, manageable, and operationally definable dimensions, concepts, and measurable variables **— 1 mark**
- Framing research objectives and formal problem formulation: Setting clear, specific, realistic, and feasible research objectives; stating research questions and framing hypotheses formally with conceptual and operational definitions **— 1 mark**

Question 17. Analyse the factors that influence a researcher's choice between primary and secondary data for a particular study. (4 Marks) [CO3, An]

- Nature, scope, and novelty of research: Unexplored phenomena or pioneering experimental forensic setups mandate tailored primary data; historical, longitudinal, or macro-demographic studies rely on secondary datasets **— 1 mark**
- Availability of resources (Time, Budget, Personnel): Primary data collection requires significant capital, laboratory infrastructure, fieldwork logistics, and time; secondary data is cost-effective, readily retrievable, and efficient **— 1 mark**
- Required precision, control, and data specificity: Primary data allows customized variable control, exact measurement protocols, and high specificity; secondary data may require extensive cleaning and adaptation **— 1 mark**
- Ethical, legal, and accessibility considerations: Availability of public archives, institutional clearance, GDPR/data privacy laws, biohazard handling permissions, and subject consent requirements **— 1 mark**

Question 18. Explain how statistical methods can support objective forensic conclusions. (4 Marks) [CO5, An]

- Quantifying evidential weight and likelihood: Replacing subjective expert impressions with quantitative metrics such as Likelihood Ratios (LR), Bayes' theorem, and Random Match Probabilities **— 1 mark**
- Distinguishing signal from random variation: Inferential statistics (t-tests, ANOVA, Chi-square) determine whether differences (e.g., glass refractive indices, soil mineral compositions, drug purities) are statistically significant or merely sampling noise **— 1 mark**
- Error rate determination and method validation: Standard deviation, confidence intervals, and ROC curves establish Limits of Detection (LOD), measurement uncertainty, false positive/negative rates adhering to Daubert legal standards **— 1 mark**

- Courtroom presentation and transparency: Provides clear, defensible, objective, and reproducible numerical boundaries that prevent cognitive bias, confirmation bias, and prosecutor's/defense fallacies **— 1 mark**

Question 19. Explain the importance of informed consent and confidentiality in research. (4 Marks) [CO1, U]

- Concept of Informed Consent: Voluntary, un-coerced agreement given by a competent participant who has been clearly briefed regarding research aims, procedures, anticipated benefits, and potential risks **— 1 mark**
- Participant autonomy and right to withdraw: Respects individual human dignity, autonomy, and provides explicit freedom to withdraw participation at any stage without penalty or prejudice **— 1 mark**
- Concept of Confidentiality: Safeguarding participant data, identity, sensitive medical/forensic records, and personal responses from unauthorized access, public disclosure, or third-party misuse **— 1 mark**
- Ethical and Legal Significance: Fosters participant trust, complies with Helsinki Declaration, Institutional Review Board (IRB) mandates, and prevents stigmatization, legal hazards, or social harm **— 1 mark**

Part C – Essay Type Questions

Answer any 1 question. Each question carries 10 marks. Maximum: 10 Marks.

Question 20. Analyse the major types of probability sampling, namely simple random, systematic, and stratified sampling. Compare their selection procedures and suitability for different population structures. (10 Marks) [CO3, An]

- Concept of Probability Sampling: Core definition wherein every unit in the sampling frame has a known, non-zero chance of selection; enables calculation of sampling error and generalizability of findings **— 2 marks**
- Simple Random Sampling (SRS): Procedure (lottery method, random number generators, equal selection probability $P = 1/N$); Suitability: Homogeneous, well-defined populations with complete sampling frames; Merits and Limitations (free from personal bias; inefficient for widely scattered or heterogeneous populations) **— 2.5 marks**
- Systematic Sampling: Procedure (selecting elements at regular intervals through the sampling frame, selecting every k-th element where $k = N/n$, with random start between 1 and k); Suitability: Ordered, sequential, or production-line datasets; Merits and Limitations (simple, rapid, even spread; high vulnerability to periodicity or hidden cyclical trends in frame) **— 2.5 marks**
- Stratified Sampling: Procedure (dividing heterogeneous population into mutually exclusive, homogeneous sub-groups/strata based on relevant variables like age, gender, geographic zone; drawing random samples proportionally or disproportionately from each stratum); Suitability: Highly heterogeneous, diverse populations; Merits and Limitations (highest representativeness, reduces sampling

error, allows sub-group analysis; requires detailed prior knowledge of population parameters) **— 3 marks**

Question 21. Analyse a given forensic dataset and select the most appropriate graphical method for its presentation, giving reasons for your choice. (10 Marks) [CO4, A]

- Principles of Forensic Data Classification and Visualisation: Identifying nature of data (qualitative/categorical vs. quantitative; continuous vs. discrete); role of graphical visualization in uncovering distributions, outliers, patterns, and trends for court and lab reporting **— 2 marks**
- Selection of Appropriate Graphical Formats by Data Type: (a) Categorical / Nominal forensic data (e.g., crime types, fingerprint pattern distribution, blood group types) -> Bar charts, segmented bar charts, or Pie charts. (b) Continuous quantitative data (e.g., glass refractive indices, blood alcohol concentrations, peak retention times) -> Histograms, Frequency Polygons, or Cumulative Frequency Ogives. (c) Bivariate/paired continuous data (e.g., height vs. long bone length, drug concentration vs. absorption peak) -> Scatter plots with regression lines. (d) Multi-group comparative distributions with outliers -> Box-and-Whisker plots **— 3 marks**
- Detailed Justification of Selected Graphical Methods: Providing technical rationale for the chosen method (e.g., why a Histogram is chosen over a Bar chart for continuous data due to contiguous intervals without artificial gaps; why a Box Plot is ideal for forensic toxicity distributions to show median, IQR, skewness, and extreme outlier data points) **— 3 marks**
- Visual Integrity, Standardization, and Admissibility in Forensic Reporting: Design best practices: clear axis labels, precise units of measurement, scales starting from zero where applicable, informative legends/titles, avoiding 3D chart distortion, and maintaining scientific objectivity to meet forensic quality standards (ISO/IEC 17025) **— 2 marks**

Marking Note: The paper instructs students to attempt at least one question from each course outcome (CO). Examiners should verify and apply Mahatma Gandhi University examination regulations regarding choice patterns and course outcome attainability when finalizing marks.